ZARISKI-LOCAL QUASI-COMPACT ETALE PRESENTATIONS VERSUS ETALE MORPHISMS WITH QUASI-COMPACT GROUPOID PROJECTIONS

THE STACKS PROJECT AUTHORS

Editorial note. No omission markers in-span.

Allowed standard prerequisites (editorial).

- Etale composition stability. Morphisms and Spaces lemmas composition-etale, composition-representable-transformations-property, morphism-schemes-gives-representable-transformation-property (cross-chapter, named). Makes U_i -> X etale in (1) implies (2).
- Images of etale morphisms are open. Properties of Spaces, lemma etale-image-open (cross-chapter, named). Defines the X_i in (2) implies (1).
- Quasi-compactness of representable morphisms. Spaces, lemma representable-morphisms-spaces-property (cross-chapter, named). Concludes U_i -> X_i quasi-compact.

Source reference. Original reference 02GN: focused-seeds/geometry/S9-decent-characterize-very-reasonable/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 02WK: focused-seeds/geometry/S9-decent-characterize-very-reasonable/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 02WJ: focused-seeds/geometry/S9-decent-characterize-very-reasonable/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 02WZ: focused-seeds/geometry/S9-decent-characterize-very-reasonable/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 06NF: focused-seeds/geometry/S9-decent-characterize-very-reasonable/ref-tags.txt, source lines 9–9. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted

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proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let X be an algebraic space over S. The following are equivalent:*

(1) *there exists a Zariski covering $X = \bigcup X_i$ and for each i a scheme U_i and a quasi-compact surjective étale morphism $U_i \to X_i$, and*
(2) *there exist schemes U_i and étale morphisms $U_i \to X$ such that the projections $U_i \times_X U_i \to U_i$ are quasi-compact and $\coprod U_i \to X$ is surjective.*

Proof. If (1) holds then the morphisms $U_i \to X_i \to X$ are étale (combine Morphisms, Lemma 02GN and Spaces, Lemmas 02WK and 02WJ). Moreover, as $U_i \times_X U_i = U_i \times_{X_i} U_i$, both projections $U_i \times_X U_i \to U_i$ are quasi-compact.

If (2) holds then let $X_i \subset X$ be the open subspace corresponding to the image of the open map $|U_i| \to |X|$, see Properties of Spaces, Lemma 06NF. The morphisms $U_i \to X_i$ are surjective. Hence $U_i \to X_i$ is surjective étale, and the projections $U_i \times_{X_i} U_i \to U_i$ are quasi-compact, because $U_i \times_{X_i} U_i = U_i \times_X U_i$. Thus by Spaces, Lemma 02WZ the morphisms $U_i \to X_i$ are quasi-compact. □